How Small Can a Reliable Prediction Set Be? Conformity-Score Design and Class-Conditional Coverage in Conformal Classification

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Abstract

Conformal prediction wraps a fixed classifier so that it returns a *set* of labels containing the true class with probability at least $1-\alpha$, without retraining and without distributional assumptions. This guarantee is only *marginal*, however: it averages over the data and does not control coverage for a particular class or a particular input. This project studies how the choice of conformity score—including a proposed temperature-scaled, distance-augmented score—affects the *size* of prediction sets at a fixed guaranteed coverage, and audits the class-conditional coverage that marginal guarantees fail to control, on MNIST and CIFAR-10 with an out-of-distribution failure audit. The ES 678 depth component derives the coverage theorem and characterises the efficiency–conditional-coverage frontier.

1 Problem and Motivation

Question. For a fixed, already-trained image classifier, how do (i) the choice of *conformity score* and (ii) the addition of feature-space *distance* information affect *both* the *size* of conformal prediction sets at a fixed, guaranteed coverage level *and* the *class-conditional* coverage that marginal guarantees do not control?

Intended prediction. For each test input x we output a *set* of labels $C(x) \subseteq \{1, \ldots, K\}$ rather than a single label, such that $P\big(Y \in C(X)\big) \geq 1 - \alpha$ without retraining and with no distributional assumptions beyond the exchangeability of calibration and test data (Vovk et al., 2005; Shafer and Vovk, 2008). A prediction is made at inference time, when no label is available and the input may be unfamiliar.

What one data row represents. One row is one image together with its ground-truth class label; features are raw pixels and the model produces $\hat{p}(y \mid x)$. We evaluate set efficiency and coverage, not raw accuracy, because a trustworthy system must know *when* it is guessing.

2 Data and Preparation

All datasets are public and load directly through `torchvision`; there are no access or sharing restrictions. One row is one image with its class label.

Dataset	Role	Size / format
MNIST	In-distribution (validity check)	60,000 train / 10,000 test, 28×28 grayscale
CIFAR-10	In-distribution (primary)	50,000 train / 10,000 test, 32×32 RGB
FashionMNIST, KMNIST	OOD for MNIST	10,000 test each
SVHN, CIFAR-100	OOD for CIFAR-10	10,000 or more test each

Access: yes (fully public, downloaded by `torchvision`); the standard, unmodified splits are used so results are comparable to the literature. **Quality checks.** We verify class balance, pixel-value ranges, and near-duplicate images. **Preprocessing.** Inputs are normalised with per-channel statistics computed on D_{train} only; CIFAR-10 augmentation (random crop, horizontal flip) is applied during training only, so no learned transformation touches the calibration or test data. Because MNIST is nearly saturated (prediction sets collapse to singletons at $\alpha \geq 0.1$), the *efficiency* study leads with CIFAR-10 and uses MNIST as a validity check.

3 Methods

Conformal construction. Let f be the classifier producing $\hat{p}(y \mid x)$, and let $s(x, y)$ be a *conformity score* (smaller = more conforming). Given the n calibration scores $S_1, \ldots, S_n$, split conformal takes the quantile

$$\hat{q} = S_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil, \tag{1}$$

where $S_{(1)} \leq \cdots \leq S_{(n)}$ are the sorted calibration scores, and returns the set

$$C(x) = \{\, y : s(x, y) \leq \hat{q} \,\}. \tag{2}$$

Under exchangeability this guarantees, for finite n,

$$1 - \alpha \;\leq\; P\big(Y \in C(X)\big) \;\leq\; 1 - \alpha + \frac{1}{n+1}. \tag{3}$$

Baselines. (1) Top-1 point prediction (accuracy reference: no set and no guarantee); (2) a simple logistic-regression classifier on MNIST, to justify the use of a CNN; and (3) the *naive* conformity score $s(x, y) = 1 - \hat{p}(y \mid x)$, which is valid but not adaptive.

Candidate scores. The first three are established; the last is ours.

$$s_{\text{naive}}(x, y) = 1 - \hat{p}(y \mid x), \tag{4}$$

$$s_{\text{APS}}(x, y) = \sum_{j < k_y} \hat{p}_{(j)}(x) + u\, \hat{p}_{(k_y)}(x), \qquad u \sim \text{Unif}(0, 1), \tag{5}$$

$$s_{\text{RAPS}}(x, y) = s_{\text{APS}}(x, y) + \lambda\, (k_y - k_{\text{reg}})^{+}, \tag{6}$$

$$s_{\gamma}(x, y) = \big(1 - \hat{p}_T(y \mid x)\big) + \gamma\, \tilde{d}_M(x, y), \qquad \gamma \geq 0, \tag{7}$$

with the temperature-scaled probabilities and the Mahalanobis distance defined by

$$\hat{p}_T(y \mid x) = \text{softmax}\big(z(x)/T\big), \tag{8}$$

$$d_M(x, y) = \sqrt{\big(\phi(x) - \mu_y\big)^{\top}\, \Sigma^{-1}\, \big(\phi(x) - \mu_y\big)}, \tag{9}$$

where $z(x)$ are the logits, $\phi(x)$ the penultimate features, and μ_y, Σ the per-class means and a shared covariance. Here k_y is the rank of y in the sorted probabilities, APS is Adaptive Prediction Sets (Romano et al., 2020), RAPS adds a rank penalty (Angelopoulos et al., 2021), T is fit by minimising the negative log-likelihood (Guo et al., 2017), and the distance term follows the Mahalanobis OOD score (Lee et al., 2018).

Where every quantity is fit (leakage discipline). The classifier is trained on D_{train}. The temperature T, the RAPS parameters λ, k_{reg}, our distance weight γ, and method selection all use D_{design} only. The class Gaussians (μ_y, Σ) and the standardisation of d_M are also fit on D_{design} (held out from training), so they are not in-sample for the classifier. The quantile $\hat{q}$ of (1) uses D_{cal} only, and D_{test} is opened once, for the final table. Mondrian (class-conditional) conformal instead calibrates a separate per-class quantile $\hat{q}_c$ from calibration points with label c, guaranteeing $P(Y \in C(X) \mid Y = c) \geq 1 - \alpha$ (Vovk, 2012). Our score (7) reduces to the temperature-scaled baseline at $\gamma = 0$, giving a clean ablation.

4 Evaluation Design

Split. A fixed, disjoint four-way split is used *before* any comparison: D_{train} (fit classifier) / D_{design} (early stopping; fit $T, \gamma, \lambda, k_{\text{reg}}$; method selection) / D_{cal} (compute $\hat{q}$ only) / D_{test} (final metrics). Model selection and tuning are thus kept entirely separate from final testing.

Metrics. We report (i) *marginal empirical coverage*, read against the finite-sample window in (3) so that a result counts as valid only if the observed coverage lies inside it; (ii) *mean*, median and 90th-percentile *set size*—the efficiency metric, because smaller sets are strictly more useful at the same error budget (Sadinle et al., 2019); (iii) singleton and empty-set rates; (iv) *class-conditional* coverage and the worst-class gap; (v) subgroup coverage by confidence decile; and (vi) coverage and size on the OOD sets. The study is repeated for $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$ and reported as mean $\pm$ standard deviation over 3–5 seeds.

5 Results and Error Analysis

Because this is a proposal, we state the expected results as testable hypotheses; a negative result is acceptable and will be reported as such.

- **H1 (validity).** Empirical marginal coverage lies in the window of (3) for every score. *Falsifier:* coverage outside the window.
- **H2 (efficiency).** APS yields a smaller mean set size than the naive score at matched coverage. *Falsifier:* naive $\leq$ APS.
- **H3 (calibration).** Temperature scaling reduces the APS set size on the miscalibrated CIFAR-10 classifier. *Falsifier:* no improvement.
- **H4 (distance term).** s_γ with $\gamma > 0$ reduces the worst-class coverage gap versus temperature-only, at a bounded cost in mean set size. *Falsifier:* the gap is unchanged or worse.
- **H5 (conditional coverage).** Mondrian attains the smallest worst-class gap but the largest mean set size. *Falsifier:* Mondrian is not best on conditional coverage.

Error analysis. We will inspect which classes and inputs break coverage, whether failures concentrate in low-confidence deciles, and how coverage and size behave on the OOD sets where exchangeability fails.

6 Graduate Depth Analysis

The primary reference discussed is the adaptive prediction sets of Romano et al. (2020), extended with rank regularisation (Angelopoulos et al., 2021), calibration (Guo et al., 2017), and a Mahalanobis feature-space distance (Lee et al., 2018). The PG depth deliverable is a study of the *efficiency–conditional-coverage frontier*, containing:

1. **Derivation / theory.** A full proof, under exchangeability, of the split-conformal coverage bound (3), with the finite-sample slack $1/(n+1)$ interpreted explicitly.
2. **Calibration / uncertainty analysis.** Class-conditional coverage per class, the worst-class coverage gap, and a comparison of the proposed s_γ against Mondrian (guaranteed) and temperature-only baselines.
3. **Sensitivity / ablation.** An $\alpha \times \gamma$ grid; the ablation $s_{\text{APS}} \to s_{\text{APS}}(T) \to s_{\text{APS}}(T) + \text{distance}$ that isolates the contribution of each term.
4. **Failure-mode audit.** Which classes and inputs break coverage; behaviour on OOD sets where exchangeability fails and the guarantee is void (see covariate-shift conformal (Tibshirani et al., 2019)); and the effect of a small calibration set n on the slack in (3).

This satisfies four of the addendum's listed options (derivation, calibration/uncertainty, sensitivity/ablation, failure-mode audit).

7 Limitations and Conclusion

One likely challenge. *Conditional coverage is not achievable for arbitrary inputs* (a known impossibility), so validity must be framed as *marginal* coverage plus an *empirical* conditional-coverage audit. A related risk is *leakage*: tuning the conformal score on the same split used to compute $\hat{q}$ breaks exchangeability and silently invalidates the guarantee—hence the strict D_{design} / D_{cal} separation above. A third risk is that the proposed distance term yields only a small or null gain; this is anticipated (H4) and will be reported honestly rather than overclaimed. **Mitigations:** separate design and calibration splits; multiple seeds with error bars; a disclosed fallback CIFAR-10 backbone; and leading with the CIFAR-10 efficiency study while treating MNIST as a validity check. **Conclusion.** By holding the classifier fixed and varying only the conformity score, the project answers a precise question: how to shrink prediction sets without losing the coverage guarantee, and where that guarantee fails conditionally.

8 Reproducibility and AI-Use Disclosure

The code is organised as a pipeline (`data`→`model`→`calibrate`→ `scores`→`conformal`→`evaluate`) with a single entry point that regenerates every figure and table from YAML configs and fixed seeds; all datasets load via `torchvision`. **Compute.** The classifier is trained once (three seeds for stability); the reported error bars vary only the calibration/test split and the randomised-APS seed, so the full grid is feasible on an Apple-silicon laptop (MPS). **Timeline (four weeks).** Week 1: data, four-way split, and a naive-conformal coverage sanity check. Week 2: APS/RAPS and temperature scaling. Week 3: the distance-augmented score plus the conditional-coverage and OOD audits. Week 4: the four-page paper, the reproduction script, and defence preparation. **Tools and AI use.** Python 3.11, PyTorch (MPS on Apple Silicon), torchvision, NumPy, SciPy, pandas, Matplotlib; LaTeX for this proposal. AI tools were used for brainstorming, boilerplate and editing; every equation, number and claim is verified independently by the author and disclosed in the final paper.

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